

# Adiba Shaira

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## Summary

Ph.D. candidate in Computer Science specializing in wearable health sensing, multimodal time-series machine learning, and Large Language Model (LLM) applications in digital health. Hands-on experience building end-to-end pipelines from raw physiological signals to interactive health insights.

## Education

**Ph.D. in Computer Science**, Stony Brook University (Candidate) Aug 2023 – Aug 2027 (expected)  
**M.S. in Computer Science**, Stony Brook University, GPA: 3.95/4.00 Dec 2025

- PiCASSo Lab; advised by Prof. Shubham Jain
- **Research Areas:** Health Sensing, Wearable Computing, Time-Series Modeling, Digital Phenotyping, Machine Learning, Large Language Models (LLMs), Explainable Machine Learning

**B.Sc. in Computer Science and Engineering**, Bangladesh University of Engineering and Technology May 2022

## Experience

**Research Assistant**, Stony Brook University May 2024 – Present

- Built LiftSafe, a machine learning system that predicts unsafe strength-training repetitions one repetition ahead from a single arm-worn Inertial Measurement Unit (IMU), reaching 87% balanced accuracy in a 24-participant study with EMG ground truth; published at IEEE/ACM CHASE 2026.
- Built CareQA, a multi-agent LLM system that answers caregivers' natural-language questions about their autistic child's sleep, activity, and physiological signals using passively sensed data from the [Simons Sleep Project](#).
- Now extending CareQA with an interactive visualization dashboard that pairs a child's sensor trends with LLM-generated explanations, and a caregiver study that grounds the system in caregivers' own questions.
- Designing passive smartphone-based sleep sensing for PTSD monitoring in a World Trade Center 9/11 first-responder cohort: extracting sleep metrics from accelerometer and screen-state data and correlating them with clinical PCL scores via longitudinal statistical modeling.

**Teaching Assistant**, Stony Brook University: CSE 373, Analysis of Algorithms Aug 2023 – May 2024

**Lecturer**, United International University, Dhaka, Bangladesh May 2022 – July 2023

- Taught Human-Computer Interaction (HCI), Analysis of Algorithms, and Computer Networks.

## Publications

**CareQA: Multi-Agent Sensemaking to Answer Caregiver Queries from Passive Sensing Data in Autism Care**  
Adiba Shaira, Michelle S. Ballan, Shubham Jain. *Under submission at IEEE PerCom 2027*.

**LiftSafe: Predicting Unsafe Repetitions in Strength Training using a Single Wearable Sensor**  
Adiba Shaira, Hunter J. Bennett, Shubham Jain. *IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (CHASE)*, 2026.

**Gene Tree Parsimony in the Presence of Gene Duplication, Loss, and Incomplete Lineage Sorting**  
Prottoy Saha, Md. Shamiul Islam, Tasnim Rahman, **Adiba Shaira**, Kazi Noshin, Rezwana Reaz, Md. Shamsuzzoha Bayzid. *RECOMB International Workshop on Comparative Genomics*, Springer Nature, pp. 110–128, 2024.

## Other Projects

**Using a Single Device to Predict Sleep Efficiency and Next-Day's Mood** 2024

Modeled next-day sleep efficiency and mood from single-wearable time-series data in the GLOBEM dataset. *Key Techniques:* Python, Pandas, NumPy, Scikit-learn, LSTM, Autoencoder models (PyTorch/TensorFlow)

**Global Disaster Visualizer** 2023

Interactive web-based visualization of global disaster data. *Key Techniques:* D3.js, JavaScript, React, Python

## Awards

**NSF Student Travel Award**, IEEE/ACM CHASE 2026

**Distinguished Travel Award**, Graduate Student Organization, Stony Brook University 2026

## Technical Skills

**Languages:** Python, Java, C/C++, SQL, JavaScript

**Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, SHAP, Large Language Models (LLMs), Multi-Agent LLM Systems, Retrieval-Augmented Generation (RAG)

**Sensing & Data:** IMU data processing, physiological signals (PPG, EDA, EEG), OpenCV, MATLAB, Pandas, NumPy

**Tools:** Git, Jupyter, Wireshark, Django, Bootstrap, MySQL, MongoDB